

DATA CLEANING PROCESS DOCUMENTATION

The dataset was cleaned and prepared to ensure accuracy, consistency, and reliability before analysis. The objective of the cleaning process was to eliminate data quality issues that could affect the validity of results and insights generated from the dataset.

CLEANING ACTIVITIES

1. Removed duplicate records to ensure that each trip was represented only once in the dataset and to prevent duplication from inflating summary statistics and KPI calculations.
2. Standardized Traffic_Level values by correcting inconsistencies in capitalization and formatting. All traffic conditions were grouped into three consistent categories: Low, Medium, and High.
3. Standardized corridor names by replacing different naming formats (e.g., *AWKA-ONITSHA*, *Awka-Onitsha*) with a single standard format: *Awka - Onitsha*.
4. Removed leading and trailing spaces from text fields such as Start_Point, Destination, Corridor, Weather, and Peak_Period to improve consistency and prevent duplicate categories during analysis.
5. Corrected capitalization inconsistencies across categorical variables to ensure uniform presentation and accurate grouping in pivot tables and visualizations.
6. Validated numerical fields including Distance_KM, Traffic_Volume, and Commute_Time_Min to confirm that values were stored in the correct format and suitable for calculations.
7. Verified date and time formats to ensure consistency across all records and support time-based analysis, filtering, and trend reporting.
8. Reviewed categorical variables such as Weather, Peak_Period, Start_Point, and Destination to ensure that values were correctly classified and free from spelling or formatting inconsistencies.
9. Conducted final validation checks to confirm that all records were complete, categories were standardized, and the dataset was suitable for reporting and visualization purposes.

OUTCOME

Following the cleaning process, the dataset became more accurate, consistent, and analysis-ready. Data quality issues such as duplicates, inconsistent naming conventions, formatting errors, and category inconsistencies were successfully addressed. The cleaned dataset provides a reliable foundation for descriptive analysis, KPI calculations, pivot table reporting, and Power BI dashboard development.

As a result, stakeholders can confidently use the dataset to evaluate commute patterns, compare traffic conditions, assess peak and off-peak travel performance, and identify factors that influence travel time along the Awka–Onitsha corridor.